

FINAL EXPLANATION: CORE MATHEMATICS GRADE 10

Student Assessment Document

FINAL EXPLANATION: CORE MATHEMATICS GRADE 10 – SUB-STRAND 1.3: QUADRATIC EXPRESSIONS AND EQUATIONS – FINAL EXPLANATION

Student Assessment Document

Student Name	_____
Class	_____
Date	_____

INSTRUCTIONS FOR STUDENTS

This is your Final Explanation for the unit. Think back to the shot put thrown at the Nairobi school sports day, the tiled courtyard, and the Kericho tea farmer's shamba. Answer each question in the space below it, in full sentences, and show all your working. In Part 1 use the height rule from Lesson 1, $h = -0.1x^2 + 1.2x + 1.5$, where h is the height in metres and x is the horizontal distance in metres. In Parts 3, 4 and 5 use the simplified height rule $h = -x^2 + 6x$, which we used in later lessons to practise factorising and solving. Support every claim with evidence from the lessons, such as your calculations, your area diagrams, your Driving Question Board entries and your parabola model. Your three-part answer to the driving question comes at the end: How can we use the mathematics of quadratic expressions and equations to predict, explain, and solve real problems?

Part 1: Why a straight line cannot describe the throw

On sports day, your classmate's slow-motion video showed the shot put tracing a smooth curve, not a straight line. (a) Using the Lesson 1 height rule $h = -0.1x^2 + 1.2x + 1.5$, work out the height of the ball when $x = 0, 2, 4, 6, 8, 10$ and 12 metres. (b) Find the first differences (change in h from one point to the next) and then the second differences. (c) Use your results to explain why a linear equation $y = mx + c$ cannot fit this curve, and say which kind of expression is needed and what feature of it produces the rise-and-fall shape.

Write your answer here:

Part 2: Forming quadratic expressions from area

Quadratic expressions arise whenever two lengths that both depend on a variable are multiplied. (a) The tiled courtyard has side lengths $(x + 2)$ and $(x + 5)$ metres. Write and expand an expression for its area using the grid/box method or FOIL, and write it in standard form $ax^2 + bx + c$. (b) Explain in one or two sentences why this area expression is quadratic and not linear. (c) The Kericho tea farmer's shamba is a rectangle with sides x and $(x + 4)$ metres. Write the area expression and expand it.

Write your answer here:

Part 3: Factorising: reversing the expansion

Factorisation reverses expansion. (a) Starting from the expanded courtyard area $x^2 + 7x + 10$, factorise it and explain how you chose your two numbers. (b) The Kericho shamba's area expression is $x^2 - 81$. Factorise it and name the identity you used. (c) Factorise the simplified shot put height expression $-x^2 + 6x$. (d) Check your answer to (a) by re-expanding.

Write your answer here:

Part 4: Solving equations to predict and decide

(a) Set the simplified shot put height rule equal to zero, $h = 0$, and solve by factorising. Explain what each solution means for the ball. (b) A Kericho tea farmer wants to fence a rectangular shamba of area 40 m^2 with one side 3 m longer than the other. Let the shorter side be x metres. Form a quadratic equation in standard form, solve it by factorisation, and state the dimensions of the shamba. (c) Explain what you do with each root and why.

Write your answer here:

Part 5: Answering the driving question

Driving question: How can we use the mathematics of quadratic expressions and equations to predict, explain, and solve real problems, from the arc of a thrown ball on our school field to the dimensions of a farmer's shamba in Kericho? Write a three-part answer. Part A, CLAIM: state your answer in 2–3 sentences. Part B, EVIDENCE: use at least two different contexts from our lessons and include at least two pieces of calculation or data. Part C, REASONING: explain how forming, factorising and solving connect to each other and to the shot put. Finally, use the perfect square identity to rewrite the simplified height expression $-x^2 + 6x$ in a form that shows the ball's highest point, and say what it tells you about the ball's flight.

Write your answer here:

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Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Use of evidence from the lessons	Uses specific data and results from the shot put, the Kericho shamba and the courtyard accurately, with correct numbers throughout.	Uses evidence from at least two contexts, with minor numerical slips that do not change the conclusion.	Gives general statements with little or no specific data, or uses numbers that do not match the lessons.
Mathematical accuracy (expanding, factorising, solving)	All expansions, factorisations and solutions are correct, shown step by step, and checked by re-expanding or substituting.	Methods are correct with one or two small errors, such as a sign slip, and working is mostly shown.	Methods are incomplete or incorrect, for example a missing middle term, wrong factor pairs or the zero-product property not used.
Reasoning and connection to the phenomenon	Clearly explains why the x^2 term gives a curve, how factorisation reverses expansion, and how the roots predict the landing point.	Explains most of the connections but one link, such as roots to landing point, is brief or unclear.	Describes procedures without explaining why they work or how they relate to the shot put.
Interpreting solutions in context	Interprets every root in context, rejects the negative length with a clear reason, and explains the vertex as the maximum height at $x = 3$ m.	Interprets the roots correctly but gives a limited reason for rejecting a solution or only a partial vertex explanation.	Gives numerical answers without units or meaning, and does not reject impossible solutions.
Communication	The Claim–Evidence–Reasoning structure is complete, uses correct	The structure is mostly complete and the vocabulary is mostly correct.	The structure is missing parts and the vocabulary is rarely used correctly.

	mathematical vocabulary, and is clearly organised.		
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ATTRIBUTION AND LICENSING

Curriculum Alignment: Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

Content & Pedagogy: Developed and adapted by the educator team at SeaVuria (registered 501(c)(3) in the United States and registered NGO in Kenya, No. 872-850A-BF11), who designed the pedagogical framework, lesson templates and exemplar lessons on which this material is based.

Production & Layout: Educational design and curriculum sequencing by SeaVuria. Technical layout, structural formatting, and digital optimization/implementation executed in partnership with ARES Education (registered 501(c)(3) in the United States and Canada (as Afretech Aid) and registered NGO in Kenya), with lesson drafting and compilation assisted by AI (Claude, Anthropic).

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